MEDIUM   ADVANCED MATH

Nonlinear functions

03

10 questions · ~15 min

SAT QUESTION BANK   2026-05-29

Q1 ADVANCED MATH

The product of a positive number x and the number that is 8 more than x is 180. What is the value of x?

A. 5

B. 10

C. 18

D. 36

Q2

ADVANCED MATH

The area A, in square centimeters, of a rectangular cutting board can be represented by the expression $ww + 9$, where w is the width, in centimeters, of the cutting board. Which expression represents the length, in centimeters, of the cutting board?

A. $ww + 9$

B. w

C. 9

D. $w + 9$

Q3 ADVANCED MATH

For the polynomial function f, the graph of $y = fx$ in the xy-plane passes through the points -50, 10, and 40. Which of the following must be a factor of $f(x)$?

A. $x + 1$

B. $x + 4$

C. x - 1

D. x - 5

Q4 GRID-IN ADVANCED MATH

$$f x = x + 6x - 4$$

If the given function f is graphed in the xy-plane, where $y = f x$, what is the x-coordinate of an x-intercept of the graph?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Time (years)	Total amount (dollars)
0	604.00
1	606.42
2	608.84

Rosa opened a savings account at a bank. The table shows the exponential relationship between the time t, in years, since Rosa opened the account and the total amount n, in dollars, in the account. If Rosa made no additional deposits or withdrawals, which of the following equations best represents the relationship between t and n?

A. $n = 1 + 604^t$

B. $n = 1 + 0.004^t$

C. $n = 6041 + 0.004^t$

D. $n = 0.0041 + 604^t$

Q6

ADVANCED MATH

$$mt = -0.0274\frac{t}{7}^2 + 7.3873\frac{t}{7} + 75.032$$

The function m gives the predicted body mass mt, in kilograms kg, of a certain animal t days after it was born in a wildlife reserve, where $t \leq 390$. Which of the following is the best interpretation of the statement "$m330$ is approximately equal to 362" in this context?

A. The predicted body mass of the animal was approximately 330 kg 362 days after it was born.

B. The predicted body mass of the animal was approximately 362 kg 330 days after it was born.

C. The predicted body mass of the animal was approximately 362 kg $\frac{330}{7}$ days after it was born.

D. The predicted body mass of the animal was approximately $\frac{330}{7}$ kg 362 days after it was born.

Q7 ADVANCED MATH

$$h(t) = -16t^2 + 110t + 72$$

The function above models the height h, in feet, of an object above ground t seconds after being launched straight up in the air. What does the number 72 represent in the function?

A. The initial height, in feet, of the object

B. The maximum height, in feet, of the object

C. The initial speed, in feet per second, of the object

D. The maximum speed, in feet per second, of the object

Q8

ADVANCED MATH

There were no jackrabbits in Australia before 1788 when 24 jackrabbits were introduced. By 1920 the population of jackrabbits had reached 10 billion. If the population had grown exponentially, this would correspond to a 16.2% increase, on average, in the population each year. Which of the following functions best models the population $p(t)$ of jackrabbits t years after 1788?

A. $p(t) = 1.162(24)^t$

B. $p(t) = 24(2)^{1.162t}$

C. $p(t) = 24(1.162)^t$

D. $p(t) = (24 \cdot 1.162)^t$

Q9 ADVANCED MATH

$$f(\theta) = -0.28(\theta - 27)^2 + 880$$

An engineer wanted to identify the best angle for a cooling fan in an engine in order to get the greatest airflow. The engineer discovered that the function above models the airflow $f(\theta)$, in cubic feet per minute, as a function of the angle of the fan θ, in degrees. According to the model, what angle, in degrees, gives the greatest airflow?

A. –0.28

B. 0.28

C. 27

D. 880

Q10 ADVANCED MATH

The function $f(w) = 6w^2$ gives the area of a rectangle, in square feet (ft^2), if its width is w ft and its length is 6 times its width. Which of the following is the best interpretation of $f(14) = 1,176$?

A. If the width of the rectangle is 14 ft, then the area of the rectangle is $1,176$ ft^2.

B. If the width of the rectangle is 14 ft, then the length of the rectangle is $1,176$ ft.

C. If the width of the rectangle is $1,176$ ft, then the length of the rectangle is 14 ft.

D. If the width of the rectangle is $1,176$ ft, then the area of the rectangle is 14 ft^2.